



College of Engineering, Construction and Living Sciences

Bachelor of Information Technology

ID511001: Programming 2

Level 5, Credits 15

Classroom Tasks

Assessment Overview

In this assessment, you will **unit test** your **Project 2 Windows Form Application** using **C#**.

Learning Outcomes

At the successful completion of this course, learners will be able to:

1. Build interactive, event-driven GUI applications using pre-built components.
2. Declare and implement user-defined classes using encapsulation, inheritance and polymorphism.

Assessments

Assessment	Weighting	Due Date	Learning Outcomes
Project 1	25%	Wednesday of Week 15 at 4.59 PM	1, 2
Project 2	35%	Sunday of Week 9 at 4.59 PM	1, 2
Theory Examination	30%	Monday of Week 16 at 4.45 PM	1, 2
Classroom Tasks	10%	Sunday of Week 9 at 4.59 PM	1, 2

Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday of Week 9 at 4.59 PM**.

Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** over all assessments in **ID511001: Programming 2**.

Authenticity

All parts of your submitted assessment **must** be completely your work. Do your best to complete this assessment without using an **AI generative tool**. You need to demonstrate to the course lecturer that you can meet the learning outcome for this assessment.

Learning to use AI tool is an important skill. While AI tools are powerful, you **must** be aware of the following:

- If you provide an AI tool with a prompt that is not refined enough, it may generate a not-so-useful response
- Do not trust the AI tool's responses blindly. You **must** still use your judgement and may need to do additional research to determine if the response is correct
- Acknowledge what AI tool you have used. In the assessment's repository **README.md** file, please include what prompt(s) you provided to the AI tool and how you used the response(s) to help you with your work

It also applies to code snippets retrieved from **StackOverflow** and **GitHub**.

Failure to do this may result in a mark of **zero** for this assessment.

Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic | Te Pūkenga** policies. Learners can view policies on the **Otago Polytechnic | Te Pūkenga** website located at <https://www.op.ac.nz/about-us/governance-and-management/policies>.

Submission

You **must** submit all application files via **GitHub Classroom**. Here is the URL to the repository you will use for your submission – <https://classroom.github.com/a/B5zbCo2J>. If you do not have not one, create a **.gitignore** and add the ignored files in this resource - <https://raw.githubusercontent.com/github/gitignore/main/VisualStudio.gitignore>. Create a branch called **classroom-tasks**. The latest application files in the **classroom-tasks** branch will be used to mark against the **Functionality** criterion. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **5:00 PM**.

Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

Resubmissions

Learners may be requested to resubmit an assessment following a rework of part/s of the original assessment. Resubmissions are to be completed within a negotiable short time frame and usually **must** be completed within the timing of the course to which the assessment relates. Resubmissions will be available to learners who have made a genuine attempt at the first assessment opportunity and achieved a **D grade (40-49%)**. The maximum grade awarded for resubmission will be **C-**.

Resits

Resits and reassessments **are not** applicable in **ID511001: Programming 2**.

Instructions

You will need to submit an application and documentation that meet the following requirements:

Functionality - Learning Outcome 1 (55%)

- Create a new **Unit Test Project (.NET Framework) Application** called **StudentManagementSystemTest**. The application needs to open without code or file structure modification in **Visual Studio**.
- You need to create **20 unit tests** covering the following:
 - All **properties** in the **Institution** class. **Three** tests are expected.
 - All **properties** in the **Person** class. **Three** tests are expected.
 - All **methods** in the **CourseAssessmentMark** class. **Seven** tests are expected.
 - The number of **Institution**, **Department** and **Course** objects after seeding. **Three** tests are expected.
 - The salary of a **Lecturer**, **Senior Lecturer**, **Principal Lecturer** and **Associate Professor**. **Four** tests are expected.

Code Quality and Best Practices - Learning Outcome 2 (40%)

- Naming Conventions (5%) - File, class, method, variable, constant, and property names should be clear, descriptive, and consistent across the codebase.
- Documentation and Comments (10%) - The code should be well-documented, with comments that explain complex logic and clarify the purpose of classes, methods or tricky sections. Using **XML** documentation format is recommended, especially for functions where the logic may not be immediately obvious to someone new to the codebase.
- Code Formatting (10%) - Consistent indentation and spacing should be used throughout the code. This includes ensuring proper indentation for code blocks, following a standard format for braces and adding blank lines between logical sections.
- Dead Code (5%) - The code should not contain any unused or redundant files, classes, methods, variables, constants, or properties. Keeping unnecessary code around can lead to confusion, clutter, and potential bugs down the line.
- Performance and Scalability (10%) - The code should be optimised for performance, using efficient algorithms and data structures to minimise bottlenecks.

Documentation and Git Usage - Learning Outcome 2 (5%)

- Provide information on how to run your unit tests in your repository **README.md** file.
- Correct spelling and grammar.
- Your **Git commit messages** should reflect the context of each functional requirement change.

Additional Information

- **Do not** rewrite your **Git** history. It is important that the course lecturer can see how you worked on your assessment over time.